INSTITUTE OF TECHNOLOGY OF CAMBODIA (ITC)

JAVA PROGRAMMING LABORATORY
REPOR

DEPARTMENT:PDI

LAB 03:Class and Object(OPP)

Course: Introduction to Programming (IP)

Group: I2-IP-Group AIECS

Submitted by:

Name: Song Panhaseth

Student ID: p20250051

- **Group:** I2-IP-

Group **Submitted to:**

Channa Bou

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PRACTICE 1

Instruction: Create a Java program using a Book class to store the Book ID, Title, Author, and Price of N books. Display all book details and calculate the average price.

LibraryMain.Java

```java
package EX1;

import java.util.ArrayList;

class Book {
    int bookID;
    String title;
    String author;
    double price;

    public Book(int BookID, String Title, String Author, double Price) {
        this.bookID = BookID;
        this.title = Title;
        this.author = Author;
        this.price = Price;
    }
    public double getPrice() {
        return price;
    }
    public void displayDetails() {
        System.out.println("ID: " + bookID + " Title: " + title + " Author:  " + author + " Price: $ " + price);
    }
}

public class libraryMain{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        ArrayList <Book> booklist = new ArrayList<>();
        System.out.print("Enter the number of books to add: ");
        int N = scanner.nextInt();
        scanner.nextLine();

        for (int i = 0; i < N; i++) {
            System.out.println("\n Enter details for Book " + (i + 1) + ":");
            System.out.print("Enter BookID: ");
            int id = scanner.nextInt();
            scanner.nextLine();
            System.out.print("Enter Title: ");
            String title = scanner.nextLine();
            System.out.print("Enter Author: ");
            String author = scanner.nextLine();
            System.out.print("Enter Price: ");
            double price = scanner.nextDouble();
            scanner.nextLine();
            booklist.add(new Book(id, title, author, price));
        }
        System.out.println("\nAll Book Details");
        double Total = 0;

        for (Book b : booklist) {
            b.displayDetails();
            Total += b.getPrice();
        }

        double average = booklist.isEmpty() ? 0 : Total / booklist.size();

        System.out.println("Total Price: $ " + Total);
        System.out.println("Average Price: $" + average);

```

Output

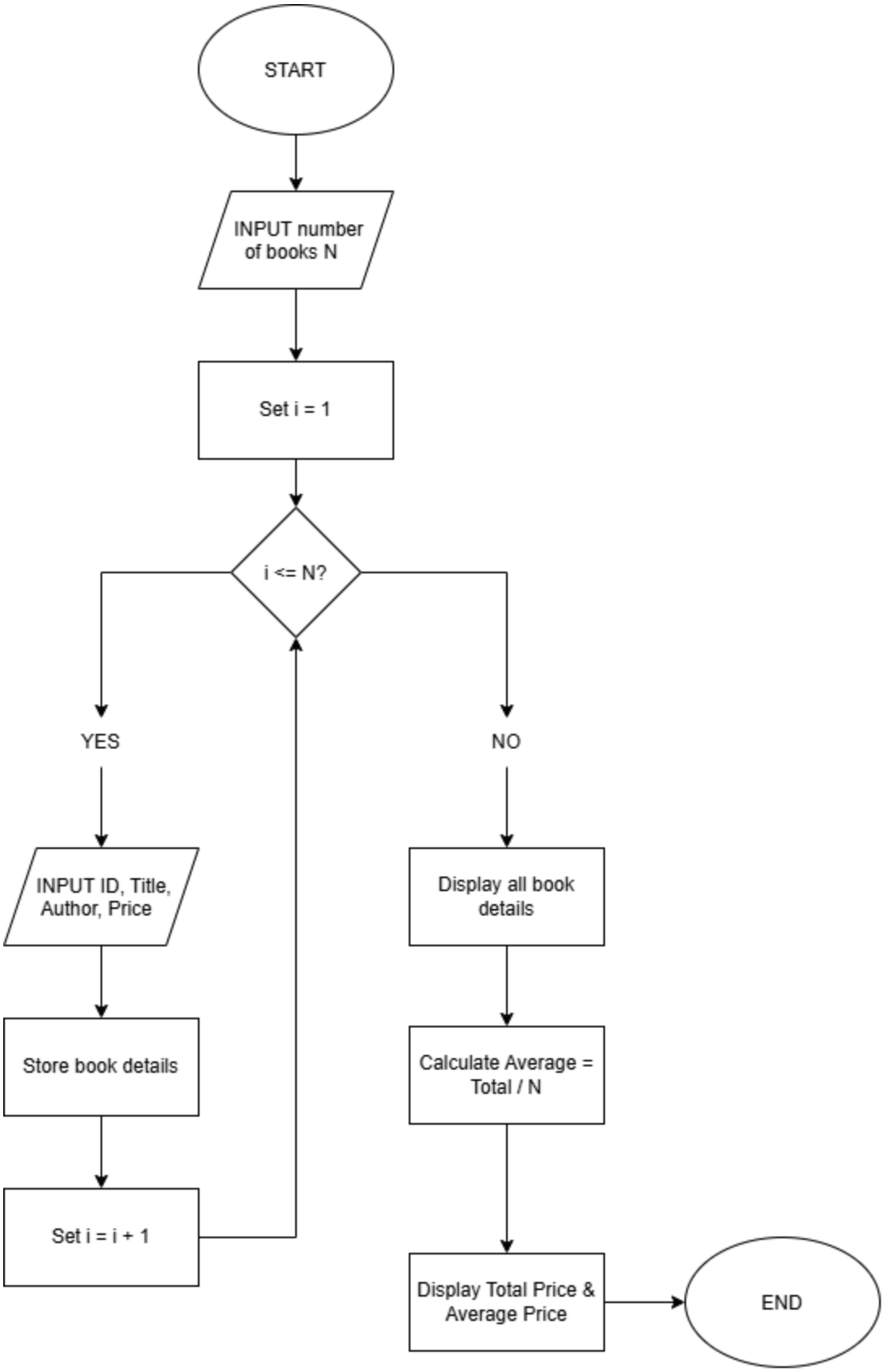
```
Enter the number of books to add: 2

 Enter details for Book 1:
Enter BookID: 3234
Enter Title: frf
Enter Author: rge
Enter Price: 23

 Enter details for Book 2:
Enter BookID: 23
Enter Title: vrf
Enter Author: re
Enter Price: 32

All Book Details
ID: 3234 Title: frf Author:  rge Price: $ 23.0
ID: 23 Title: vrf Author:  re Price: $ 32.0
Total Price: $ 55.0
Average Price: $27.5
```

Flowchart


START
INPUT number of books N
Set i = 1
i <= N?
YES
NO
INPUT ID, Title, Author, Price
Store book details
Set i = i + 1
Display all book details
Calculate Average = Total / N
Display Total Price & Average Price
END

Pseudocdoe

```
START
INPUT N
CREATE Book arraylist of size N

FOR i = 0 TO N-1
    INPUT Book ID
    INPUT Title
    INPUT Author
    INPUT Price
    STORE Book details into ArrayList
END FOR

FOR each Book in ArrayList
    DISPLAY Book Details (ID, Title, Author, Price)
    ADD Price to Total
END FOR

CALCULATE Average = Total / N
DISPLAY Total Price, Average Price

END
```

Practice2

Instruction: Create a Java program using a Temperature class to store the Celsius temperatures of N cities. Convert each temperature to Fahrenheit and display both Celsius and Fahrenheit values.

Ex2.Java

```java
package EX2;

import java.util.Scanner;

class Temperature {
    double celsius;
    double fahrenheit;

    void setTemperature(double c) {
        celsius = c;
    }
    void convert() {
        fahrenheit = (celsius * 9.0 / 5.0) + 32;
    }
    void display() {
        System.out.println("Celsius: " + celsius);
        System.out.println("Fahrenheit: " + fahrenheit);
    }
}
public class Ex2 {
        public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the number of cities: ");
        int N = input.nextInt();

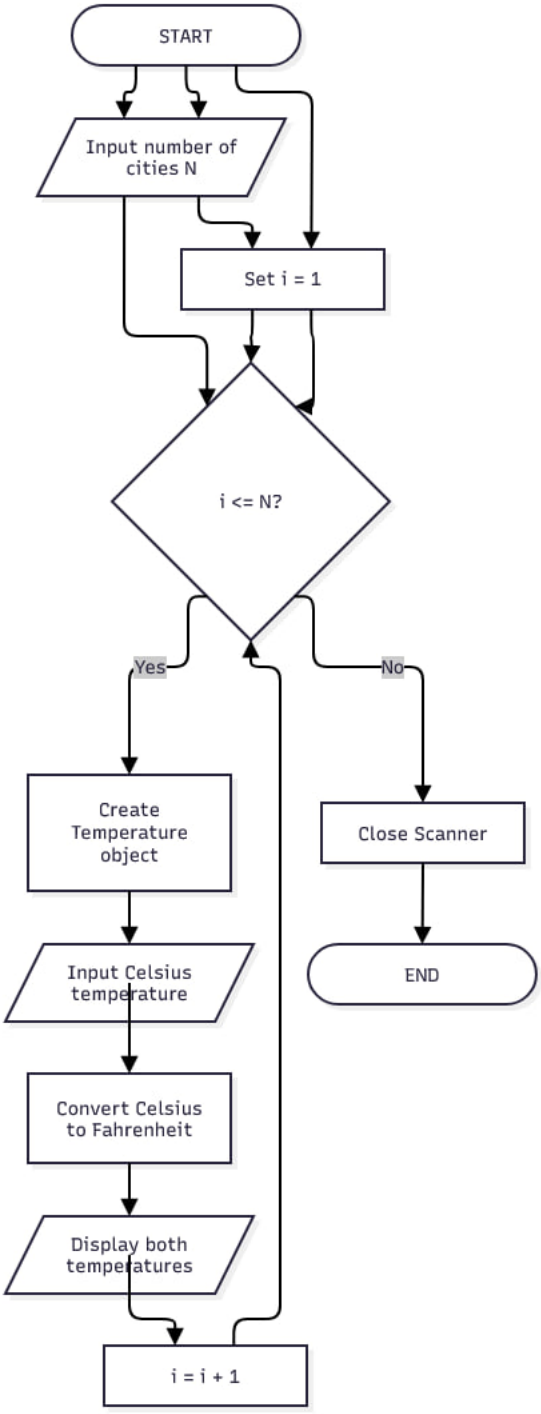
        for (int i = 1; i <= N; i++) {
            Temperature temp = new Temperature();

            System.out.print("Enter Celsius temperature for city "
                    + i + ": ");
            double c = input.nextDouble();

            temp.setTemperature(c);
            temp.convert();

            System.out.println("City " + i);
            temp.display();
            System.out.println();
        }
        input.close();
    }
}
```

Flowchart2


START
Input number of cities N
Set i = 1
i <= N?
Yes
No
Create Temperature object
Input Celsius temperature
Convert Celsius to Fahrenheit
Display both temperatures
i = i + 1
Close Scanner
END

Pseudocode

```
CLASS Temperature
    DECLARE celsius, fahrenheit AS REAL
    METHOD setTemperature(c)
        celsius = c
    END METHOD
    METHOD convert()
        fahrenheit = (celsius * 9 / 5) + 32
    END METHOD
    METHOD display()
        OUTPUT "Celsius: ", celsius
        OUTPUT "Fahrenheit: ", fahrenheit
    END METHOD
END CLASS
MAIN
    DECLARE N, i AS INTEGER
    DECLARE c AS REAL

    OUTPUT "Enter the number of cities: "
    INPUT N

    FOR i = 1 TO N
        CREATE OBJECT temp OF Temperature
        OUTPUT "Enter Celsius temperature for city ", i
        INPUT c
        CALL temp.setTemperature(c)
        CALL temp.convert()
        OUTPUT "City ", i
        CALL temp.display()
    END FOR
END MAIN
```

Practice3

Instruction: Create a Java program using an Employee class to store the Employee ID, Name, and Basic Salary of N employees. Calculate HRA (20%), TCA (10%), and Gross Salary, then display all employee details.

Practice3.java

```java
package EX3;
import java.util.Scanner;

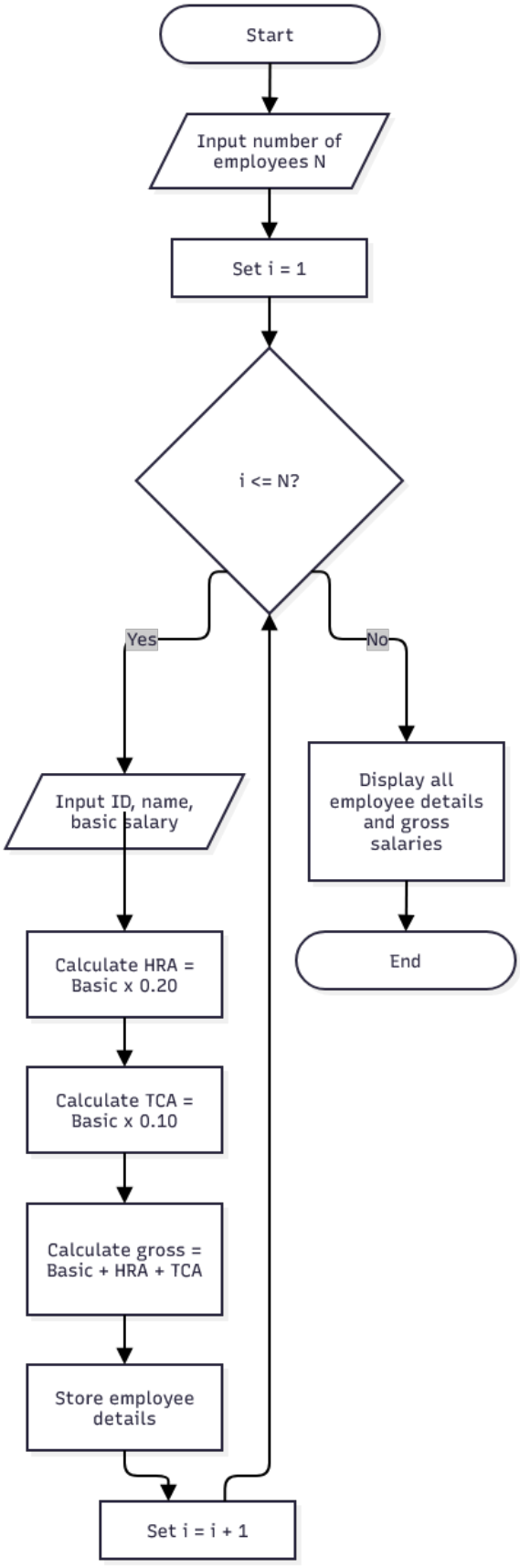
class Employee {
    int id;
    String name;
    double basicSalary, hra, tca, grossSalary;
    Employee(int i, String n, double s) {
        id = i;
        name = n;
        basicSalary = s;
        hra = s * 0.20;
        tca = s * 0.10;
        grossSalary = s + hra + tca;
    }
    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("TCA: " + tca);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("-----------------");
    }
}
public class practice3 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter number of employees: ");
        int n = input.nextInt();
        input.nextLine();
        Employee[] emp = new Employee[n];
        for (int i = 0; i < n; i++) {
            System.out.print("Enter ID: ");
            int id = input.nextInt();
            input.nextLine();
            System.out.print("Enter name: ");
            String name = input.nextLine();
            System.out.print("Enter basic salary: ");
            double salary = input.nextDouble();
            input.nextLine();
            emp[i] = new Employee(id, name, salary);
        }
        System.out.println("\nEmployee Salary Details:");
        for (int i = 0; i < n; i++) {
            emp[i].display();
        }
        input.close();
    }}
```

Output

```
Enter number of employees: 2
Enter ID: 32
Enter name: ref
Enter basic salary: 23
Enter ID: 4323
Enter name: erfe
Enter basic salary: 3224

Employee Salary Details:
ID: 32
Name: ref
Basic Salary: 23.0
HRA: 4.600000000000005
TCA: 2.3000000000000003
Gross Salary: 29.900000000000002
------------------
ID: 4323
Name: erfe
Basic Salary: 3224.0
HRA: 644.8000000000001
TCA: 322.40000000000003
Gross Salary: 4191.2
------------------
```

Flowchart


Start
Input number of employees N
Set i = 1
i <= N?
Yes
No
Input ID, name, basic salary
Calculate HRA = Basic x 0.20
Calculate TCA = Basic x 0.10
Calculate gross = Basic + HRA + TCA
Store employee details
Set i = i + 1
Display all employee details and gross salaries
End

Pseudocode

```
START
INPUT N
CREATE Employee array of size N

FOR i = 0 TO N-1
    INPUT ID
    INPUT Name
    INPUT Basic Salary

    HRA = Basic Salary * 0.20
    TCA = Basic Salary * 0.10
    Gross Salary = Basic Salary + HRA + TCA

    STORE employee details
END FOR

FOR i = 0 TO N-1
    DISPLAY ID, Name, Basic Salary
    DISPLAY HRA, TCA, Gross Salary
END FOR

END
```

Practice4

Instruction: Create a Java program using a Student class to store the Student ID, Name, and marks of 5 subjects for N students. Calculate the total marks and average, assign a grade based on the average, and display all student details.

Java

Student1

```java
package EX4;

public class Student1 {

    private String name, ID;
    private float[] scores = new float[5];
    private float sum = 0.0f, average=0.0f;
    //Student constructor with params accepting the student name, ID, and an array of scores.
    Student1(String name, String ID, float[] scores){
        this.name = name;
        this.ID = ID;
        this.scores = scores;
    }
    //Getter functions for name, ID, and scores alongside the sum and avg.
    public String nameGetter(){
        return this.name;
    }
    public String IDGetter(){
        return this.ID;
    }
    public float scoresGetter(int index){
        return this.scores[index];
    }
    public float sumGetter(){
        return this.sum;
    }
    public float averageGetter(){
        return this.average;
    }
    //Honestly just added this to make the code alot easier to read.
    public void calcSumAndAverage(){
        for(float i : this.scores){
            this.sum += i;
        }
        this.average = this.sum/5.0f;
    }
    //Grade determiner function.
    public char gradeDetermine(){
        if(this.average >= 90.0f){
            return 'A';
        }else if (this.average >= 75.0f){
            return 'B';
        }else if (this.average >= 50.0f){
            return 'C';
        }else{
            return 'F';
        }
    }
}
```

StudentGrade

```java
package EX4;

import java.util.Scanner;
import java.util.ArrayList;
import java.util.List;

public class StudentGrade{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String name, ID;
        String[] subject = {"Mathematics", "Physics", "Chemistry", "Biology", "English"};
        int numberOfStudents;
        List<Student1> students = new ArrayList<>();

        System.out.print("Enter the amount of students: ");
        numberOfStudents = sc.nextInt();
        sc.nextLine();
        for(int i = 0; i < numberOfStudents; i++){
            //Accept information of each student.
            System.out.println("+----------------------------------------------+");
            System.out.print("Enter name: ");
            name = sc.nextLine();
            System.out.print("Enter ID: ");
            ID = sc.nextLine();
            float[] scores = new float[5];
            for(int j = 0; j < 5; j++){
                System.out.printf("Enter your score for %s: ", subject[j]);
                scores[j] = sc.nextFloat();
            }
            sc.nextLine();
            //Adds a new student into the list. and calculates sum and avg.
            students.add(new Student1(name, ID, scores));
            students.get(i).calcSumAndAverage();
        }
        System.out.println("END==========================================\n");
        System.out.println("Evaluating results...");
        System.err.println("Printing...");
        for(int i = 0; i < numberOfStudents; i++){
            System.out.println("\n==========================================");
            System.out.printf("%s (%s), here are your scores: \n",students.get(i).nameGetter(),students.get(i).IDGetter());
            for(int j = 0; j < 5; j++){
                System.out.printf("\t%s: %.2f\n", subject[j], students.get(i).scoresGetter(j));
            }
            System.out.printf("\tTotal Marks: %.2f\n", students.get(i).sumGetter());
            System.out.printf("\tAverage: %.2f\n", students.get(i).averageGetter());
            System.out.printf("\tYour letter grade: %c\n", students.get(i).gradeDetermine());
            System.out.println("==========================================");
        }
        sc.close();
    }
```

Output

```
Enter the amount of students: 3
+----------------------------------------------+
Enter name: de
Enter ID: 232
Enter your score for Mathematics: 423
Enter your score for Physics: 423
Enter your score for Chemistry: 423
Enter your score for Biology: 2323
Enter your score for English: 32
+----------------------------------------------+
Enter name: rfege
Enter ID: 434
Enter your score for Mathematics: 324
Enter your score for Physics: 53
Enter your score for Chemistry: 45
Enter your score for Biology: 32
Enter your score for English: 43
+----------------------------------------------+
Enter name: fr
Enter ID: 43
Enter your score for Mathematics: 53
Enter your score for Physics: 34
Enter your score for Chemistry: 23
Enter your score for Biology: 43
Enter your score for English: 32
END==========================================

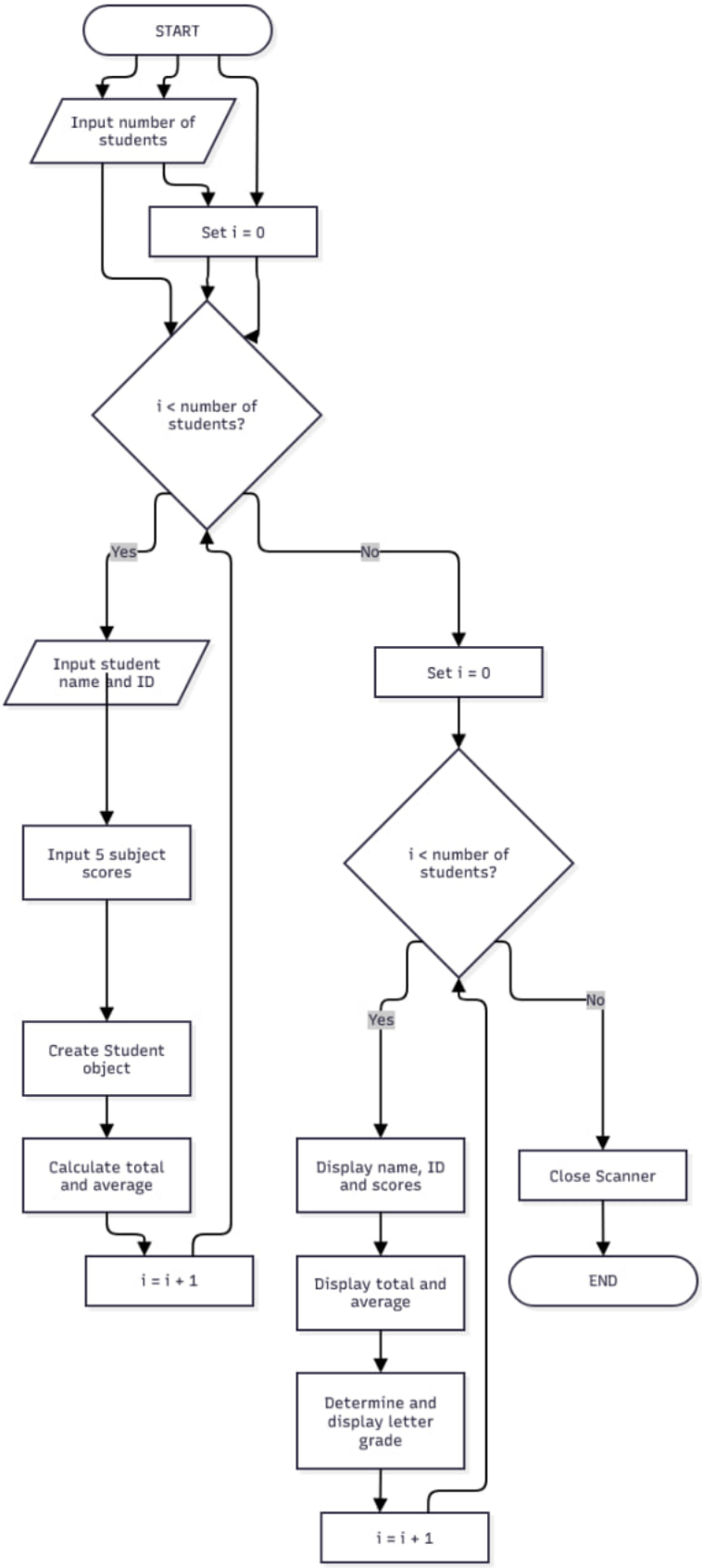
Evaluating results...
Printing...
```

```
==========================================
de (232), here are your scores:
        Mathematics: 423.00
        Physics: 423.00
        Chemistry: 423.00
        Biology: 2323.00
        English: 32.00
        Total Marks: 3624.00
        Average: 724.80
        Your letter grade: A
==========================================

==========================================
rfege (434), here are your scores:
        Mathematics: 324.00
        Physics: 53.00
        Chemistry: 45.00
        Biology: 32.00
        English: 43.00
        Total Marks: 497.00
        Average: 99.40
        Your letter grade: A
==========================================

==========================================
fr (43), here are your scores:
        Mathematics: 53.00
        Physics: 34.00
        Chemistry: 23.00
        Biology: 43.00
        English: 32.00
        Total Marks: 185.00
        Average: 37.00
        Your letter grade: F
==========================================
```

Flowchart


START
Input number of students
Set i = 0
i < number of students?
Yes
No
Input student name and ID
Input 5 subject scores
Create Student object
Calculate total and average
i = i + 1
Set i = 0
i < number of students?
Yes
No
Display name, ID and scores
Display total and average
Determine and display letter grade
i = i + 1
Close Scanner
END

Pseudocode

```
START

INPUT numberOfStudents

FOR each student

    INPUT student name
    INPUT student ID

    FOR 5 subjects
        INPUT subject score
    END FOR

    CREATE Student object

    CALCULATE total marks
    CALCULATE average

END FOR

FOR each student

    DISPLAY student name and ID
    DISPLAY 5 subject scores
    DISPLAY total marks
    DISPLAY average

    IF average >= 90
        DISPLAY Grade A
    ELSE IF average >= 75
        DISPLAY Grade B
    ELSE IF average >= 50
        DISPLAY Grade C
    ELSE
        DISPLAY Grade F
    END IF

END FOR

CLOSE Scanner

END
```

Practice5

Instruction: Create a Java program using an Account class to store the account number, name, and balance of N customers. Allow users to deposit and withdraw money while checking sufficient balance, then display the updated account details.

Java

Account1

```java
package EX5;

public class Account1 {
    public String accNum;
    private float balance = 0.0f;
    private String name;

    Account1(String accNum, String name, float balance){
        this.accNum = accNum;
        this.name = name;
        this.balance = balance;
    }
    public String accNumGetter(){
        return this.accNum;
    }
    public float balGetter(){
        return this.balance;
    }
    public String nameGetter(){
        return this.name;
    }
    public void withdraw(float amount){
        if(amount > this.balance){
            System.out.println("Withdrawal amount exceeded current balance.");
        } else {
            this.balance -= amount;
            System.out.printf("\nWithdrew %.2f from %s (%s) successfully!\n", amount, this.name, this.accNum);
        }
    }
    public void deposit(float amount){
        this.balance += amount;
    }
}
```

BankingSystem1

```java
package EX5;

import java.util.Scanner;
import java.util.List;
import java.util.ArrayList;

public class BankingSystem1 {

    public static Account1 findAccount(List<Account1> accounts, String accNum){
        for(Account1 acc : accounts){
            if (acc.accNumGetter().equals(accNum)){
                return acc;
            }
        }
        return null;
    }

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);
        List<Account1> accounts = new ArrayList<>();
        System.out.print("Welcome to our Banking System! Enter number of customers: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            System.out.println("\nCustomer #" + (i + 1));
            String accNo;
            while (true) {
                System.out.println("+-----------------------------------------------------+");
                System.out.print("Account Number (must be 14 digits long): ");
                accNo = sc.nextLine().trim();
                if (accNo.matches("\\d{14}")) break; //Java doesn't have .size :(
                System.out.println("Invalid! Must be exactly 14 digits.");
            }
            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("Balance: ");
            float balance = sc.nextFloat();
            sc.nextLine();
            accounts.add(new Account1(accNo, name, balance));
        }

        int choice;
        do {
            System.out.println("\n-------------------- BANK  MENU --------------------");
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Display all accounts");
            System.out.println("0. Exit");
            System.out.println("-----------------------------------------------------");
            System.out.print("Choose: ");
            choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1 || choice == 2) {
                System.out.print("Enter account number: ");
                String accNo = sc.nextLine().trim();
                Account1 acc = findAccount(accounts, accNo);

                if (acc == null) {
                    System.out.println("Account not found, double check the entered Account Number.");
                    continue;
                }

                System.out.print("Enter amount: ");
                float amount = sc.nextFloat();


                if (choice == 1) acc.deposit(amount);
                else acc.withdraw(amount);
                sc.nextLine();
                System.out.println("-----------------------------------------------------");
                System.out.println("Updated account details:");
                System.out.printf("\tUsername: %s\n", acc.nameGetter());
                System.out.printf("\tAccount Number: %s\n", acc.accNumGetter());
                System.out.printf("\tCurrent Balance: %.2f\n", acc.balGetter());
                System.out.println("-----------------------------------------------------");
            } else if (choice == 3) {
                for (Account1 acc : accounts){
                    System.out.println("-----------------------------------------------------");
                    System.out.printf("\tUsername: %s\n", acc.nameGetter());
                    System.out.printf("\tAccount Number: %s\n", acc.accNumGetter());
                    System.out.printf("\tCurrent Balance: %.2f\n", acc.balGetter());
                    System.out.println("-----------------------------------------------------");
                }
            }
        } while (choice != 0);

        System.out.println("Thank you for using our banking system!");
        System.out.println("Exiting...");
        sc.close();
    }
}
```

Output

```
Welcome to our Banking System! Enter number of customers: 2

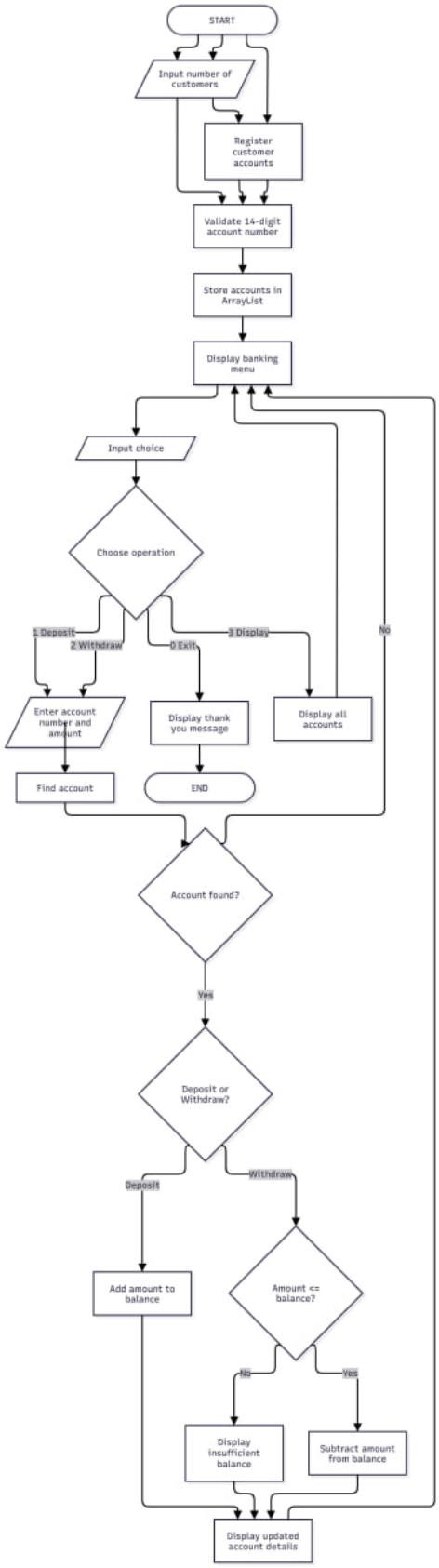
Customer #1
+-----------------------------------------------------+
Account Number (must be 14 digits long): 23
Invalid! Must be exactly 14 digits.
+-----------------------------------------------------+
Account Number (must be 14 digits long): 12345678901234
Name: ah
Balance: 34

Customer #2
+-----------------------------------------------------+
Account Number (must be 14 digits long): 09876543211234
Name: fd
Balance: 32

-------------------- BANK  MENU --------------------
1. Deposit
2. Withdraw
3. Display all accounts
0. Exit
-----------------------------------------------------
Choose: 2
Enter account number: 32
Account not found, double check the entered Account Number.

-------------------- BANK  MENU --------------------
1. Deposit
2. Withdraw
3. Display all accounts
0. Exit
-----------------------------------------------------
Choose: 0
Thank you for using our banking system!
Exiting...
```

Flowchart


START
Input number of customers
Register customer accounts
Validate 14-digit account number
Store accounts in ArrayList
Display banking menu
Input choice
Choose operation
1 Deposit
2 Withdraw
0 Exit
3 Display
No
Enter account number and amount
Display thank you message
Display all accounts
Find account
END
Account found?
Yes
Deposit or Withdraw?
Deposit
Withdraw
Add amount to balance
Amount <= balance?
No
Yes
Display insufficient balance
Subtract amount from balance
Display updated account details

Pseudocode

```
START

INPUT number of customers

FOR each customer
    INPUT account number

    WHILE account number is not 14 digits
        DISPLAY invalid account number
        INPUT account number again
    END WHILE

    INPUT customer name
    INPUT initial balance

    CREATE Account object
    ADD account to account list
END FOR

DO
    DISPLAY banking menu
    INPUT choice

    IF choice = 1 OR choice = 2 THEN

        INPUT account number
        SEARCH for account

        IF account not found THEN
            DISPLAY "Account not found"

        ELSE
            INPUT amount

            IF choice = 1 THEN
                DEPOSIT amount
            ELSE
                WITHDRAW amount
                IF amount > balance THEN
                    DISPLAY "Insufficient balance"
                END IF
            END IF

            DISPLAY updated account details
        END IF

    ELSE IF choice = 3 THEN
        DISPLAY all accounts
    END IF

WHILE choice != 0
DISPLAY thank you message
CLOSE Scanner
END
```